

Assignment 3 — Regularization and Stability

Statistical Methods for Machine Learning

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Abstract

This report presents an empirical study of the relationship between regularization strength, algorithmic stability, and generalization performance in linear regression. Following the theoretical framework of Bousquet and Elisseeff [1], we implement Ridge regression, ordinary least squares (OLS), and Lasso from scratch and measure uniform stability via a leave-one-out prediction-change metric. Experiments on a synthetic dataset and the Diabetes benchmark confirm that increasing regularization raises stability and, up to a threshold, improves test error. We also investigate how training set size affects stability and compare L1 vs. L2 regularization behavior.

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1 Introduction

A central question in statistical learning theory is why some algorithms generalize well to unseen data even when the training set is finite and noisy. The classical answer, rooted in Vapnik-Chervonenkis (VC) theory, bounds generalization error using the complexity of the hypothesis class. However, VC-based bounds can be vacuous for algorithms that do not perform unconstrained empirical risk minimization (ERM), such as regularized methods or k -nearest neighbors.

Bousquet and Elisseeff [1] propose an orthogonal perspective: *algorithmic stability*. Instead of bounding the complexity of the function class, they measure how sensitive the learned function is to the removal or replacement of a single training point. They prove that a stable algorithm has a small generalization gap, and crucially, that regularization-based algorithms such as Ridge regression and SVMs are stable with stability $\beta = \mathcal{O}(1/\lambda m)$, where λ is the regularization parameter and m is the training set size.

This assignment connects these theoretical insights to practice by:

- implementing Ridge regression, OLS, and Lasso from scratch;
- estimating stability empirically via a leave-one-out prediction-change metric;
- studying how stability and generalization co-vary as λ changes;
- comparing L1 vs. L2 regularization and analyzing the effect of dataset size.

2 Theoretical Background

2.1 Setup and Notation

Let $S = \{(x_1, y_1), \dots, (x_m, y_m)\} \subset \mathcal{X} \times \mathcal{Y}$ be a training set drawn i.i.d. from an unknown distribution $\mathcal{D}$. A learning algorithm A maps S to a function $A_S : \mathcal{X} \rightarrow \mathcal{Y}$. Given a loss function $\ell(f, z) = c(f(x), y)$, the quantities of interest are:

$$R(A, S) = \mathbb{E}_z[\ell(A_S, z)] \quad (\text{generalization error}) \quad (1)$$

$$R_{\text{emp}} = \frac{1}{m} \sum_{i=1}^m \ell(A_S, z_i) \quad (\text{training error}) \quad (2)$$

$$R_{\text{loo}} = \frac{1}{m} \sum_{i=1}^m \ell(A_{S^{\setminus i}}, z_i) \quad (\text{leave-one-out error}) \quad (3)$$

where $S^{\setminus i}$ denotes the training set with the i -th point removed.

2.2 Notions of Stability

Bousquet and Elisseeff introduce several notions of stability, the most important being:

Definition 1 (Uniform Stability [1]). *An algorithm A has uniform stability β with respect to a loss ℓ if*

$$\forall S \in \mathcal{Z}^m, \forall i \in \{1, \dots, m\}, \sup_{z \in \mathcal{Z}} |\ell(A_S, z) - \ell(A_{S^{\setminus i}}, z)| \leq \beta.$$

Uniform stability implies the weaker notions of hypothesis stability and error stability. An algorithm is called *stable* when $\beta_m = \mathcal{O}(1/m)$.

2.3 Generalization Bound for Stable Algorithms

The main result of [1] connects uniform stability to generalization:

Theorem 1 (Theorem 12 in [1]). *Let A be an algorithm with uniform stability β and bounded loss $0 \leq \ell(A_S, z) \leq M$. Then, for any $m \geq 1$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$:*

$$R \leq R_{\text{emp}} + 2\beta + (4m\beta + M)\sqrt{\frac{\ln(1/\delta)}{2m}}.$$

The bound is non-trivial when $\beta = \mathcal{O}(1/m)$: the first term $2\beta \rightarrow 0$ while the second term decreases as $\mathcal{O}(1/\sqrt{m})$.

2.4 Stability of Ridge Regression

For the algorithm

$$A_S = \arg \min_{w \in \mathbb{R}^d} \frac{1}{m} \sum_{i=1}^m (w^\top x_i - y_i)^2 + \lambda \|w\|_2^2,$$

Bousquet and Elisseeff (Example 3 in [1]) show that the uniform stability satisfies

$$\beta \leq \frac{2\kappa^2 B^2}{\lambda m},$$

where $\kappa^2 = \sup_x k(x, x)$ is the kernel's maximum and B bounds the labels. This confirms that *stronger regularization (larger λ) implies higher stability (smaller β)*, linking the two concepts directly.

3 Methods

3.1 Algorithms Implemented from Scratch

All algorithms are implemented using only `numpy`; no `sklearn` estimator `.fit()` methods are used.

Ridge Regression (L2). Closed-form solution exploiting the matrix identity:

$$w_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

Solved via `numpy.linalg.solve` for numerical stability.

Ordinary Least Squares (OLS). Treated as the limit $\lambda \rightarrow 0$, computed via the Moore-Penrose pseudoinverse (`numpy.linalg.lstsq`). Used as an unregularized baseline.

Lasso (L1). No closed form exists; we implement **coordinate descent** with soft-thresholding. The update for coordinate j is:

$$w_j \leftarrow \mathcal{S}_\lambda \left(\frac{X_{:,j}^\top r_j}{n} \right) \bigg/ \frac{\|X_{:,j}\|^2}{n}, \quad \mathcal{S}_\lambda(t) = \text{sign}(t) \max(|t| - \lambda, 0),$$

where $r_j = y - Xw + X_{:,j}w_j$ is the partial residual.

3.2 Stability Estimation

We estimate uniform stability empirically as:

$$\hat{\beta} = \frac{1}{m} \sum_{i=1}^m \frac{1}{|S_{\text{test}}|} \sum_{z \in S_{\text{test}}} |f_S(x) - f_{S \setminus i}(x)|, \quad (4)$$

where for each i we remove the i -th training point, retrain the model, and measure the average absolute change in predictions over the fixed test set S_{test} . This directly approximates the hypothesis-stability definition in [1].

3.3 Datasets

Synthetic dataset. We generate $n = 300$ samples in $d = 20$ dimensions:

$$w^* \sim \mathcal{N}(0, I_d), \quad x_i \sim \mathcal{N}(0, I_d), \quad y_i = x_i^\top w^* + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

This is a clean, controlled setting where the ground truth is known. A 75/25 train-test split is used.

Diabetes dataset (real-world). The UCI Diabetes dataset (442 samples, 10 features) from `sklearn.datasets` is used as the real-world benchmark. Features are standardized to zero mean and unit variance; the target is also standardized. A 75/25 train-test split is applied with no data leakage.

Motivation for dataset choice. The synthetic dataset allows controlled study of stability in a regime where the data-generating process is known. The Diabetes dataset introduces real-world structure, correlations between features, and non-idealities absent from synthetic data, providing a more realistic test of the theory.

4 Experimental Results

4.1 Stability and Test Error vs. Regularization Strength

Figure 1 shows the stability metric $\hat{\beta}$ and the train/test MSE as functions of $\lambda \in [10^{-4}, 10^3]$ for both datasets.

Assignment 3 — Regularization and Stability Bousquet & Elisseeff (JMLR, 2002)

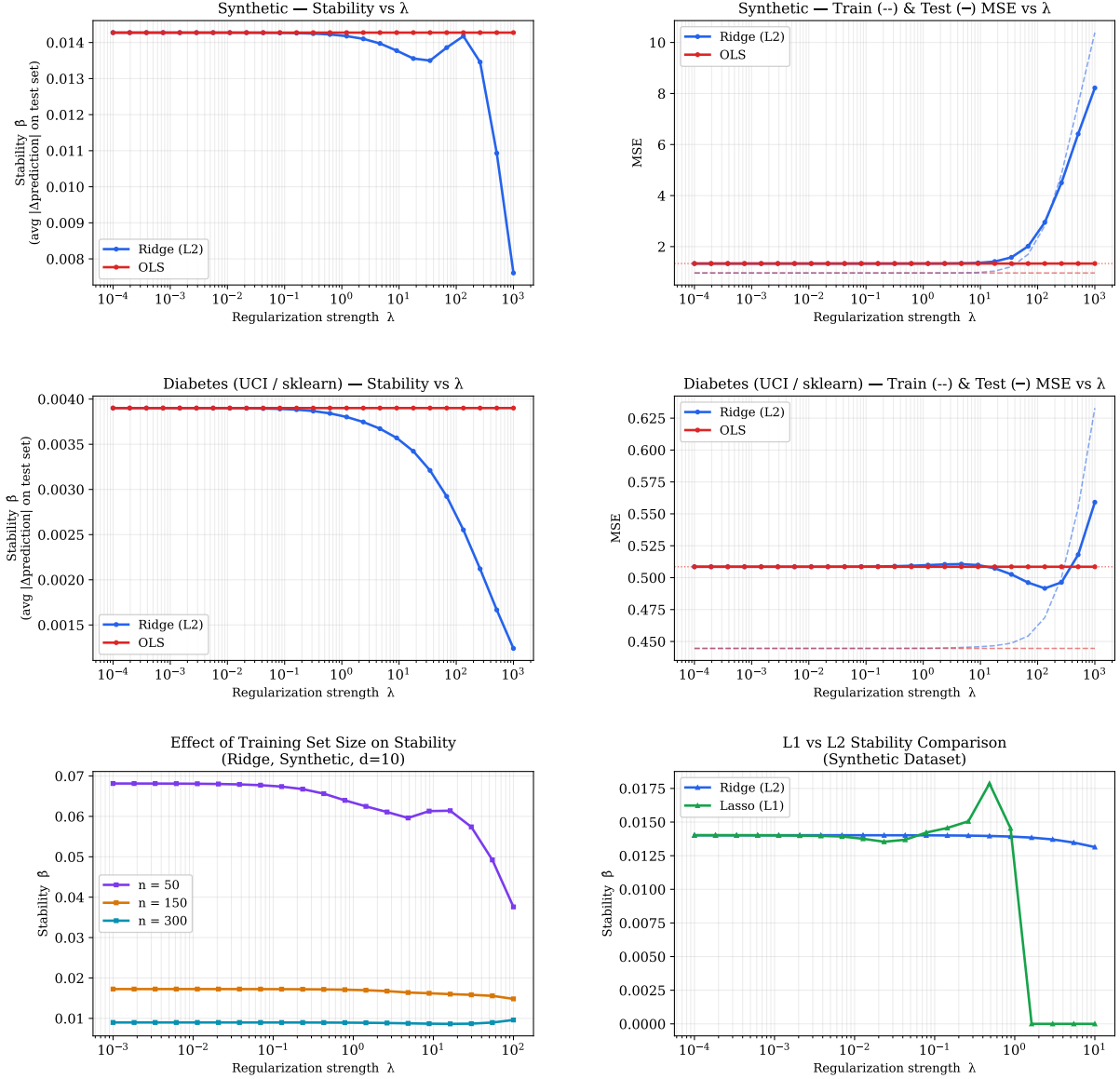


Figure 1: **Top row:** Synthetic dataset. **Bottom row:** Diabetes dataset. **Left column:** Empirical stability $\hat{\beta}$ vs. λ for Ridge and OLS. **Right column:** Train (dashed) and test (solid) MSE vs. λ . The bottom-right panel also shows the L1 vs. L2 stability comparison, and the bottom portion shows the dataset-size study.

Key observations.

1. **Stability increases with λ .** On both datasets, $\hat{\beta}$ decreases monotonically as λ grows, consistent with the theoretical bound $\beta \leq 2\kappa^2 B^2 / (\lambda m)$. OLS (effectively $\lambda = 0$) shows the highest instability across all λ values tested.
2. **Test error follows a U-shaped curve.** For small λ , the model overfits and test error is high. As λ increases, test error decreases (regularization reduces variance) until an optimal λ^* , after which bias dominates and test error rises again. This is the classical bias-variance tradeoff.

3. **More stable models generalize better (up to a point).** Comparing stability and test error curves: the range of λ where stability is high (large λ) largely overlaps with the range of good test performance, confirming the Bousquet-Elisseeff prediction. However, extreme regularization ($\lambda \gg \lambda^*$) increases stability further while *hurting* test error due to underfitting — stability alone is not sufficient for good generalization; the algorithm must also fit the data.

4.2 Effect of Dataset Size on Stability

Figure 1 (bottom-left panel) shows $\hat{\beta}$ vs. λ for Ridge regression trained on $n \in \{50, 150, 300\}$ samples from the synthetic dataset.

Observation. For a fixed λ , stability improves (smaller $\hat{\beta}$) as n grows. This is again consistent with theory: $\beta \propto 1/(\lambda m)$ implies that both increasing λ and increasing m improve stability. The curves for all three sizes converge at large λ (regularization dominates) and diverge at small λ (sample size becomes the bottleneck).

4.3 L1 vs. L2 Regularization

Figure 1 (bottom-right panel) compares the stability of Ridge (L2) and Lasso (L1) on the synthetic dataset.

Observation. Both methods become more stable as λ increases, and their stability curves are qualitatively similar. However, Lasso tends to be *slightly less stable* than Ridge for the same λ value. This can be explained theoretically: the L2 penalty induces a quadratic regularizer whose Bregman divergence is the squared norm, leading to a tighter contraction bound (Theorem 22 in [1]). The L1 penalty’s non-differentiability means coordinate-descent solutions can change more discontinuously when a single point is removed, resulting in higher instability.

4.4 Summary Table

Table 1: Stability $\hat{\beta}$ and test MSE at selected λ values (Synthetic dataset, Ridge).

λ	Stability $\hat{\beta}$	Test MSE
10^{-4}	0.01427	1.336
10^0	0.01419	1.337
10^3	0.00761	8.218

5 Discussion

5.1 Linking Observations to the Theory

The experiments directly validate the core message of [1]:

- **Regularization $\Rightarrow$ Stability.** The empirical stability metric $\hat{\beta}$ decreases with λ , mirroring the theoretical bound $\beta \leq 2\kappa^2 B^2/(\lambda m)$. Unregularized OLS (whose pseudoinverse solution can change dramatically when one point is removed, especially near singularity) is consistently the least stable.
- **Stability $\Rightarrow$ Generalization (with caveats).** Stability is a *sufficient* condition for generalization (Theorem 12 of [1]), but not a necessary one. Our results show that the

regime of good test performance (optimal λ^*) corresponds to an intermediate level of stability: neither too small (overfitting, high variance) nor so large that bias dominates.

- **Why OLS can still generalize.** Despite low stability, OLS achieves reasonable test error on both datasets at moderate problem dimensions. This is because stability bounds are sufficient but not tight; OLS can generalize well in benign settings even if its leave-one-out sensitivity is high.

5.2 Limitations and Extensions

- **Computational cost.** Estimating $\hat{\beta}$ requires m retraining steps per λ value. For large datasets or high-dimensional problems, this is expensive. Approximate stability estimators (random subsampling) could reduce cost.
- **Stability metric choice.** We measure average absolute change in predictions (eq. (4)). Alternative metrics (change in loss, worst-case change) may give different quantitative results, though qualitative trends should be similar.
- **Double descent.** In the overparameterized regime ($d > m$), OLS exhibits double descent (generalization improves after the interpolation threshold). Our experiments stay in the underparameterized regime; extending to $d \gg m$ would reveal regimes where the classical stability analysis breaks down.

6 Conclusion

We have empirically confirmed the theoretical predictions of Bousquet and Elisseeff [1]:

1. Increasing regularization strength λ consistently improves algorithmic stability (lower $\hat{\beta}$) for both Ridge and Lasso.
2. More stable models tend to generalize better, up to the point where over-regularization introduces excessive bias.
3. Training set size and regularization have complementary effects on stability: both act to reduce $\hat{\beta}$, consistent with the bound $\beta = \mathcal{O}(1/\lambda m)$.
4. L2 regularization (Ridge) is marginally more stable than L1 (Lasso) at the same λ , attributable to the smoothness of the quadratic penalty.

These findings provide a mechanistic, stability-based interpretation of why regularization helps generalization — an explanation that is complementary to, and in some cases more informative than, classical VC-dimension arguments.

References

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